

Troubleshooting Network Connectivity — Guided Notes

Unit 5.2.2 | CompTIA Network+ N10-009 Obj. 5.2 | N10-008 5.2

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

A workstation has IP address 169.254.45.3. This tells you the client _____ to contact a DHCP server. The client self-assigned an _____ address. From this state, the client can communicate with _____ but nothing else.

Question 2

Two devices have the same IP address on the same VLAN. Describe the symptom a user would experience and name the command on the affected client that might reveal a changing MAC address for the gateway.

Question 3

Complete the four-step diagnostic sequence for 'I can't get to the internet.'

Step 1: `ping \_\_\_\_\_` — tests local Layer 3 connectivity.

Step 2: `ping \_\_\_\_\_` — tests internet path by IP, bypassing DNS.

Step 3: `nslookup \_\_\_\_\_` — tests DNS resolution.

Step 4: If Step 2 works but Step 3 fails, the problem is _____.

Question 4

A client can ping the gateway and external IPs but cannot reach port 443 on an internal server.

`Test-NetConnection 10.1.50.10 -Port 443` returns a timeout. What are two possible causes?

Question 5

A VLAN that exists on an access switch but is not added to the _____ between that switch and the distribution layer will have Layer 2 connectivity locally but cannot reach the _____. Clients will appear to have valid IPs but go nowhere.

Question 6

NAT misconfiguration can cause internal hosts to send packets to the internet with _____ source addresses that are dropped by ISP routers. A new subnet that wasn't added to the NAT _____ list is the most common cause after a VLAN expansion.

Question 7

Multiple users on one switch cannot reach the internet, but users on other switches are fine. The uplink from that switch shows _____ in `show interface`. What is the most efficient next step to narrow the cause?

Question 8 — Real World Application

REAL WORLD: A teacher moves to a new classroom. Their laptop has the correct IP, gateway, and DNS. They can ping the gateway. They can ping 8.8.8.8. But no websites load. Nslookup for any public domain returns the correct IP. Describe what you would check next and why.

Download the answer key from the class server:

```
wget http://[SERVER]/resources/network-engineer/5-2-2-troubleshooting-network-connectivity/5-2-2-troubleshooting-network-connectivity-answer-key.pdf
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